جامعة لوسيل
Lusail University
QATAR

College of Information Technology

# Project

# Smart AI Assistant with Human Feedback (Multimodal)

## Objective

In this project, you will build a simplified multimodal AI system that generates text using a Large Language Model (LLM) and images using a diffusion model. The system will incorporate human feedback to evaluate and improve the generated text, simulating the concept of Reinforcement Learning from Human Feedback (RLHF), while demonstrating how modern AI systems integrate multiple modalities.

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## Project Description

You will develop an AI assistant that:

**PART 1:**

1. Takes a **user prompt**
2. Generates multiple **text responses** using an LLM
3. Ranks and Improves text using **human feedback (RLHF simulation)**

**PART 2:**

4. Generates an **image from the same prompt** using a diffusion model

**Note: You are not required to train models from scratch. You may use pretrained models and existing libraries.**

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## Project Requirements

### Part 1 — Text Generation with RLHF

#### ◆ Step 1: Generate Text

- Use a pretrained LLM (e.g., GPT-2) **via Hugging Face (Transformers library)**
- Input: user prompt
- Output: **2–3 different responses**

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#### ◆ Step 2: Human Feedback

- The user should:
  - Select the best response OR
  - Rate responses (1–5)

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#### ◆ Step 3: Ranking System

- Store feedback
- Rank responses based on:
  - Number of votes OR
  - Average rating

This simulates a **reward model in RLHF**

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### Part 2 — Image Generation (Diffusion Model)

- Use a diffusion model (e.g., Stable Diffusion) **via Hugging Face (Diffusers library)**
- Input the **same user prompt used in Part 1**
- Output: at least **1 generated image**

No feedback is required for images

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## Prompt Examples (Guidelines)

To ensure consistency between text and image generation, choose prompts that can be **both described in words and visualized as an image**.

You must use the **same prompt** suitable for:

- Text generation (Part 1)
- Image generation (Part 2)

### Prompt Examples:

- “A futuristic smart city with flying cars”
- “A cat wearing sunglasses on the beach”
- “An astronaut walking on Mars at sunset”
- “A robot helping humans in daily life”
- “A modern university campus in the future”
- ……..

### ❌ Prompts to Avoid

- “Explain machine learning”
- “Define artificial intelligence”
- “Summarize this paragraph”

These prompts work for text but are not suitable for image generation.

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## Deliverables

- **Source Code**
    - Python script or Google Colab notebook
- **Presentation (8–12 slides)**
  Include:
    - System architecture
    - Models used:
        - GPT-2 (LLM)
        - Stable Diffusion (Diffusion model)
    - Explanation of RLHF simulation
    - Description of implementation
    - Challenges and observations
- **Demo (3–5 minutes)**
    - Show text generation and feedback process
    - Show image generation

## Grading and Evaluation

The project constitutes **20%** of the total course grade, distributed as follows:

| | |
|---|---|
| Text Generation (LLM) | 4 |
| RLHF Simulation (Text Feedback) | 4 |
| Image Generation (Diffusion) | 4 |
| Functionality and Code Quality | 4 |
| Presentation and Questions | 4 |
| **Total** | **20points** |

## Important dates:

- **Slides submission:** **20/04/2026** (by email)
- **Project Presentations:** Starting from 22/04/2026 (during regular classes)